Lian Cataytay

Data Analyst & Economist · Analytics, dashboards, and econometric research

BUSINESS & INTELLIGENCE · MACHINE LEARNING & RISK · ECONOMICS & RESEARCH

The analyst who thinks like an economist

A data analyst and economist from Davao City, working where business intelligence, machine learning, and econometric research meet.

Lian is a data analyst and economist from the Philippines. His work spans econometrics, business intelligence, and machine learning. He has built projects involving retention analytics, AI-powered churn prediction, banking risk analysis, and workforce intelligence.

Data is only valuable when it leads to better decisions. Many organizations already have the numbers; what they lack is a clear way to interpret them. That is the gap he focuses on closing.

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BY THE NUMBERS

3+
years in analytics and research

13
end-to-end projects

12
with public code on GitHub

1.30
CGPA, BS Economics, University of Southeastern Philippines

TOOLKIT

Python · SQL · R · Stata · Excel · Power BI · Tableau · Pandas · Scikit-learn · SHAP

CONTACT

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Curriculum vitae

EXPERIENCE

Aug 2025 – Oct 2025

Data Analyst Intern
Socio-Economic Research and Data Analytics Center

Standardized multi-source demographic data in RStudio and Excel, built charts and Excel dashboards, and turned statistical outputs into briefs for policymakers and program staff.

Oct 2024 – Present

Freelance Data Analyst
Self-employed

Builds revenue and expense trackers with budget-vs-actual checks, KPI dashboards covering 8+ marketing and finance metrics, and regression-based scenario and A/B lift models.

Aug 2024 – Jan 2026

Econometrics and Applied Economics Research Analyst
University of Southeastern Philippines

Built Prais-Winsten, time-series, Logit, and Probit models for macroeconomic, inflation, and loan-delinquency research, and synthesized the findings for policy audiences.

Aug 2022 – Aug 2024

Lead Feature Editor
The Collegiate Headlight, USeP

Wrote, edited, and quality-checked 100+ published pieces and mentored junior writers on evidence use and narrative structure.

Aug 2020 – Apr 2022

Administrative Assistant
Saint John Marie Vianney Academy of Sulop, Inc.

Managed accounting logs, records, and reporting, and produced 50+ educational and promotional materials.

EDUCATION

University of Southeastern Philippines

BS Economics, major in Development Economics
Davao City · CGPA 1.30

Coursework in econometrics, time-series analysis, financial economics, development economics, statistics, and project evaluation.

TOOLKIT

Python · SQL · R · Stata · Excel · Power BI · Tableau · Pandas · Scikit-learn · SHAP

LEAD PROJECT · PUBLIC HEALTH ANALYTICS · FORECASTING · RISK SCORING

EpiWatch Mindanao Public Health Risk Forecasting

A simulated public health early warning system with case forecasting, anomaly detection, risk scoring, and dashboard-ready outputs.

Context. Public health teams need a structured way to monitor disease trends, detect abnormal increases, and prioritize municipalities for validation and preparedness review.

Method. I generated synthetic surveillance data, engineered incidence and lag features, forecasted next-four-week case burden, detected anomalies against local baselines, and built an Early Warning Risk Index.

Findings. The prototype produced 3,024 municipality-week records and classified the latest week into 1 High Risk, 1 Alert, 14 Watch, and 8 Normal municipalities. Jolo ranked highest with a 0.80 risk index after a sharp rise above its baseline, and the forecasting model's MAE was 36.76 cases.

Tools: Python, Pandas, NumPy, Excel, Power BI-ready CSV
Code: github.com/liancataytayjk-coder/epiwatch-mindanao

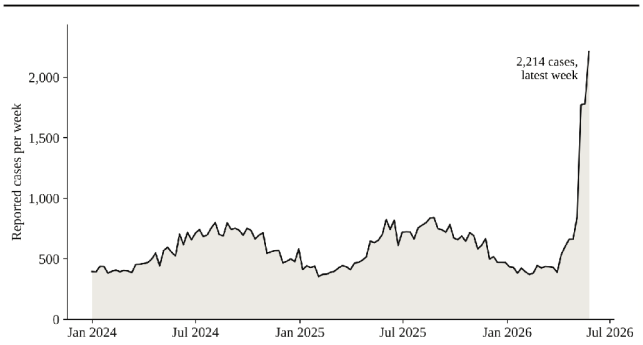


Figure 1. Weekly reported cases across 24 Mindanao municipalities, January 2024 to May 2026. *Synthetic dengue-like surveillance data. Source: EpiWatch Mindanao repository.*

BUSINESS & INTELLIGENCE · CUSTOMER ANALYTICS · RETENTION · SEGMENTATION

E-commerce Customer Retention Analytics

Customer behavior analysis focused on retention, segmentation, cohort patterns, and churn-risk insights.

Context. I wanted to understand which customers drive the most value and which groups are most at risk of churn.

Method. I used Python, Pandas, cohort analysis, customer segmentation, and Tableau dashboarding.

Findings. Across 93,358 customers and $19.9M in revenue, the churn rate was 59.1%. Champions made up only 6.9% of customers while 14.0% were at risk, showing where retention effort should go first.

Tools: Python, Pandas, Tableau, Cohort analysis
Code: github.com/liancataytayjk-coder/ecommerce-customer-retention-analytics

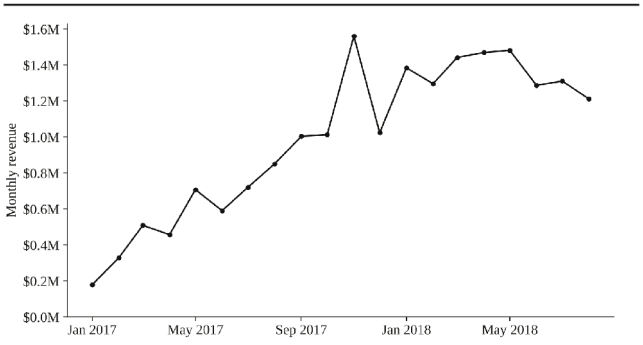


Figure 2. Monthly marketplace revenue, 2017 to 2018. *Months with fewer than 100 orders are omitted. Olist Brazilian e-commerce data. Source: E-commerce Customer Retention Analytics repository.*

BUSINESS & INTELLIGENCE · MARKETPLACE ADS · ACOS/TACOS · BUDGET ALLOCATION

eCommerce Marketplace Ads Performance Dashboard

Marketplace ads reporting with ACoS/TACoS KPIs, slow-moving SKU analysis, data quality checks, and budget recommendations.

Context. Marketplace advertising teams often manage separate exports from Amazon Ads, Google Ads, TikTok Ads, and eBay Ads, making it harder to compare spend, revenue, profit, and inventory impact.

Method. I consolidated platform schemas, cleaned SKU and campaign fields, joined product and inventory data, calculated ACoS/TACoS-style KPIs, and built validation checks for reporting quality.

Findings. The workflow processed 42,360 ad rows and 6,500 sales rows, generated $86.7M in attributed ad revenue, produced a 6.80 average ROAS, and flagged 4,903 campaign/SKU rows for reduction or pause.

Tools: Excel, SQL, Python, Streamlit
Code: github.com/liancataytayjk-coder/ecommerce-marketplace-ads-performance-dashboard

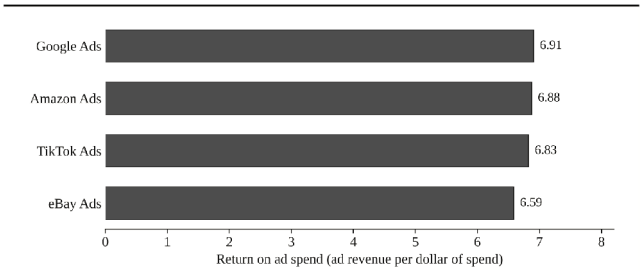


Figure 3. Return on ad spend by marketplace platform. *Simulated ad exports across four platforms. Source: eCommerce Marketplace Ads Performance Dashboard repository.*

BUSINESS & INTELLIGENCE · FP&A · FORECASTING · VARIANCE ANALYSIS

FP&A Revenue, Cost & Expense Forecasting Dashboard

Monthly FP&A reporting with budget variance, forecast accuracy, KPI trends, scenario analysis, and anomaly review.

Context. FP&A teams need reliable monthly reporting to understand revenue trends, cost movement, expense control, forecast accuracy, and KPI performance.

Method. I standardized messy finance data, calculated actual vs budget and actual vs forecast variances, built scenario forecasts, tracked KPI movement, and flagged anomalies.

Findings. The workflow analyzed $281.6M in revenue, 50.8% gross margin, 98.2% forecast accuracy, 71 unfavorable variances, and 369 anomaly records for review.

Tools: Excel, SQL, Python, Streamlit
Code: github.com/liancataytayjk-coder/fpa-revenue-cost-expense-forecasting-dashboard

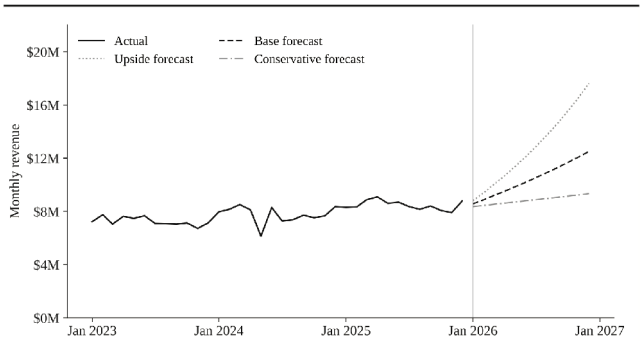


Figure 4. Monthly revenue, 2023 to 2025, with 2026 forecast scenarios. *Synthetic financials. Source: FP&A Revenue, Cost & Expense Forecasting Dashboard repository.*

BUSINESS & INTELLIGENCE · CRM HYGIENE · EVENT MARKETING · SALES OPERATIONS

CRM & Event Marketing Data Hygiene System

CRM cleaning, duplicate resolution, contact validation, event follow-up, and marketing-ready outputs.

Context. Marketing and sales teams often work from CRM exports with duplicate contacts, invalid emails, inconsistent phone numbers, and incomplete event records.

Method. I standardized contact fields, scored record completeness, resolved duplicate groups, matched event registrations, and prepared post-event follow-up outputs.

Findings. The workflow cleaned 1,500 raw CRM records into 1,187 unique contacts, removed 313 duplicates, flagged 469 invalid emails, and prepared 310 referrals for follow-up.

Tools: Excel, Power Query logic, Python, Pandas
Code: github.com/liancataytayjk-coder/crm-event-marketing-data-hygiene-system

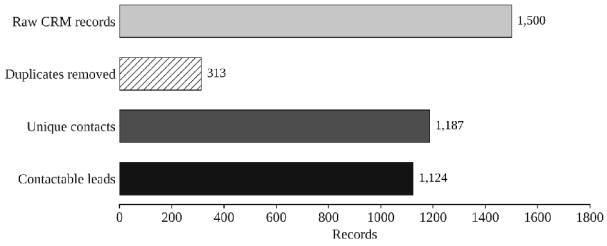


Figure 5. From raw export to contactable leads. *469 invalid emails and 283 invalid phone numbers were also flagged. Source: CRM & Event Marketing Data Hygiene System repository.*

BUSINESS & INTELLIGENCE · HR ANALYTICS · ATTRITION RISK · WORKFORCE INTELLIGENCE

Employee Attrition Risk and Workforce Intelligence Analysis

A workforce intelligence project identifying employee groups and teams with stronger attrition-risk patterns.

Context. I explored how workforce data could be used to identify employees and teams with higher attrition risk.

Method. I used Python, Pandas, attrition-risk scoring, and hotspot analysis.

Findings. The project identified teams and employee groups showing stronger retention risk patterns.

Tools: Python, Pandas, Risk scoring, Hotspot analysis
Code: not public



Figure 6. Key figures from the workforce attrition analysis. *Summary figures as reported in the project; the underlying HR data are not public.*

MACHINE LEARNING & RISK · MACHINE LEARNING · EXPLAINABILITY · CHURN RISK

AI Customer Churn Intelligence System

An explainable churn-prediction workflow using machine learning, SHAP, and retention-focused reporting.

Context. I wanted to build a churn-prediction workflow that was not only accurate, but also explainable.

Method. I applied machine-learning classification models, SHAP explainability, and Power BI reporting.

Findings. Among the 1,409 customers the model scored, those in the High Risk segment churned at 78.8%, against 12.9% in the Low Risk segment. Contract type, pricing, and service usage were the strongest drivers, and $27,972 in revenue was flagged as at risk.

Tools: Python, scikit-learn, SHAP, Power BI
Code: github.com/liancataytayjk-coder/ai-customer-churn-intelligence-system

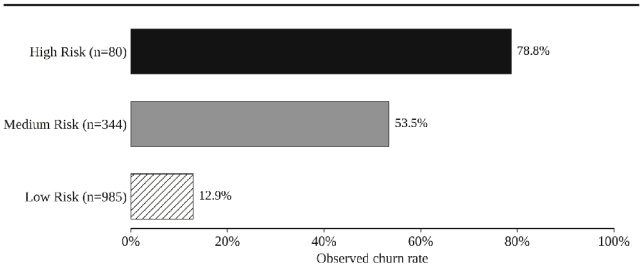


Figure 7. Observed churn by model risk segment. *1,409 customers scored by the model. Source: AI Customer Churn Intelligence System repository.*

MACHINE LEARNING & RISK · BANKING ANALYTICS · CREDIT RISK · FINANCIAL PERFORMANCE

Banking Risk and Financial Performance Analysis

A financial-risk project identifying borrower and loan characteristics associated with default exposure.

Context. I wanted to understand which borrower and loan characteristics are most associated with default risk.

Method. I used Python, Pandas, a classification model, and feature-importance analysis, then grouped loans into five equal risk bands and estimated expected-loss exposure.

Findings. The model reached 0.72 ROC AUC. Loans in the Very High risk band defaulted at 53%, against 2% in the Very Low band, and that band carried 53% of expected-loss exposure. Credit score and interest rate were the strongest drivers.

Tools: Python, Pandas, Classification analytics, Feature importance
Code: github.com/liancataytayjk-coder/banking-risk-and-financial-performance-analysis

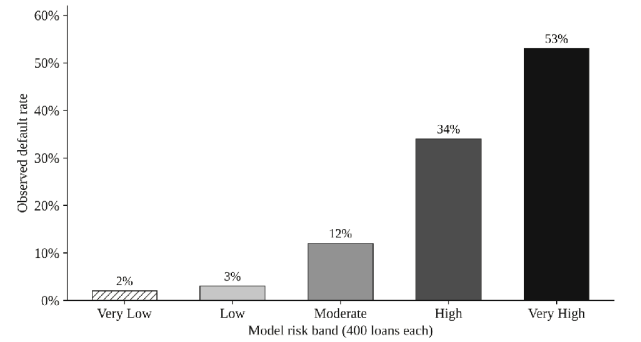


Figure 8. Observed default rate by model risk band. *Synthetic 2,000-loan portfolio; overall default rate 20.8%. Source: Banking Risk and Financial Performance Analysis repository.*

MACHINE LEARNING & RISK · NLP · SENTIMENT ANALYSIS · BUSINESS INTELLIGENCE

Customer Sentiment Intelligence Dashboard

Sentiment analysis of customer reviews and business-facing metrics for interpreting customer pain points and satisfaction trends.

Context. I wanted to see how raw customer reviews could be turned into something decision-makers can actually use.

Method. I cleaned and processed the data in Python, applied sentiment-analysis workflows, and prepared dashboard-ready KPIs in Power BI.

Findings. Of the 500 reviews, 41.6% were positive, 33.0% neutral, and 25.4% negative, for an NPS of 30. Coffee quality, staff behavior, and amenities were the most frequent themes, and high-risk reviews were flagged for follow-up.

Tools: Python, Power BI, Pandas, NLP workflow
Code: github.com/liancataytayjk-coder/customer-sentiment-intelligence-dashboard

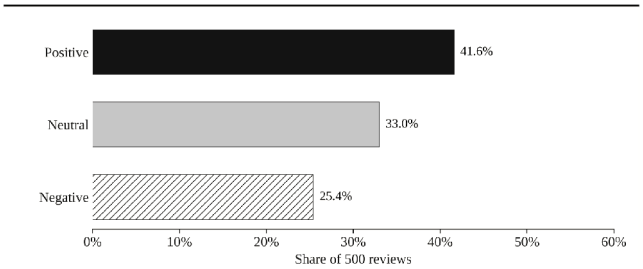


Figure 9. Sentiment of 500 multilingual customer reviews. *Net Promoter Score of 30. Source: Customer Sentiment Intelligence Dashboard repository.*

ECONOMICS & RESEARCH · AGRICULTURAL ECONOMICS · SYNTHETIC CONTROL METHOD · IMPACT EVALUATION

CARP and Agricultural Productivity: A Synthetic Control Study

A synthetic control evaluation of how the 1988 agrarian reform program affected Philippine labor productivity, land productivity, and TFP.

Context. The Comprehensive Agrarian Reform Program (CARP), launched in 1988, is central to Philippine debates on equity and rural development, but its effect on national agricultural productivity is rarely tested against a credible counterfactual.

Method. I used the synthetic control method (SCM) to build a "synthetic Philippines" that matches pre-1988 productivity trends, then compared post-reform labor productivity, land productivity, and TFP. A GDP per capita placebo checked whether the gaps were specific to agriculture.

Findings. From 1989 to 2008, Philippine agriculture ran below its synthetic control: labor productivity was about 18.2% lower, land productivity 16.1% lower, and TFP about 9% lower. GDP per capita showed only a 2.5% gap, so the shortfall is specific to agriculture and driven mainly by slower TFP growth.

Tools: R, R Markdown, Synthetic control, TFP analysis
Code: github.com/liancataytayjk-coder/agricultural-scm-productivity-analysis

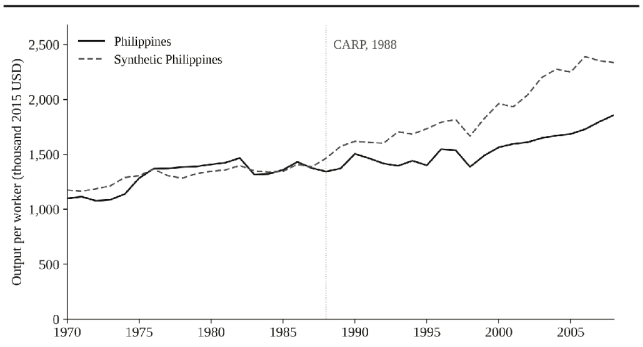


Figure 10. Labor productivity: the Philippines and its synthetic control, 1970 to 2008. *After 1988 the Philippines averages about 18.2% lower than its synthetic control (1989 to 2008). Source: Cataytay, Deluna and Evangelio, synthetic control analysis.*

ECONOMICS & RESEARCH · MACROECONOMICS · MIGRATION · ARDL MODELING

Philippine Emigration ARDL Macroeconomic Analysis

An econometric analysis of how macroeconomic conditions relate to Philippine emigration trends.

Context. I explored how macroeconomic conditions relate to Philippine emigration trends.

Method. I used Stata, ARDL bounds testing, ECM modeling, and time-series diagnostics.

Findings. Five of seven ARDL specifications showed a long-run relationship. In the baseline model, a 1% rise in Philippine GDP per capita is linked to a 0.46% fall in long-run emigration, and error-correction terms of −0.56 to −1.00 mean most deviations close within a year. COVID-19 produced a large negative level shift.

Tools: Stata, ARDL bounds testing, ECM modeling, Time-series diagnostics
Code: github.com/liancataytayjk-coder/philippine-emigration-ardl-macroeconomic-analysis

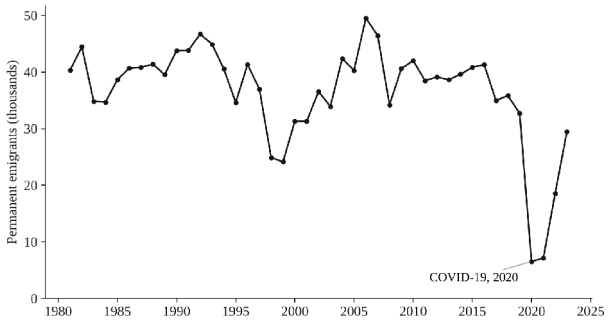


Figure 11. Permanent Philippine emigrants, 1981 to 2023. *Permanent emigrant counts. Source: project repository.*

ECONOMICS & RESEARCH · LABOR ECONOMICS · TIME-SERIES TRENDS · POLICY INSIGHT

Philippines Labor Market Trend Analysis

A labor-market analysis of employment quality, workforce composition, and underemployment patterns in the Philippines.

Context. I wanted to better understand employment and labor-market trends in the Philippines.

Method. I used Python, Pandas, and time-series trend analysis.

Findings. In 2025, labor force participation stood at 64.1%, unemployment at 4.2%, and underemployment at 11.4%. Underemployment remained almost three times the unemployment rate, pointing to job quality rather than job availability as the bigger gap.

Tools: Python, Pandas, Time-series analysis, Excel
Code: github.com/liancataytayjk-coder/philippines-labor-market-trend-analysis

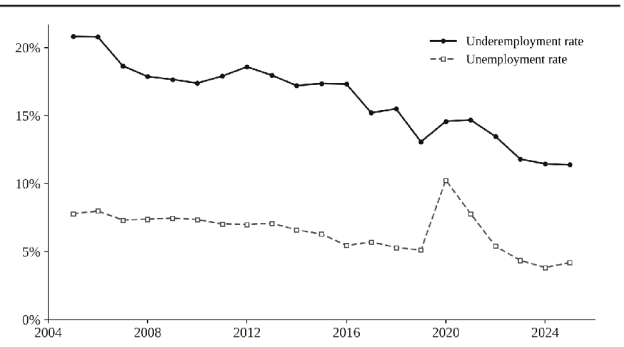


Figure 12. Unemployment and underemployment rates, 2005 to 2025. *Annual figures from Labor Force Survey indicators. Source: project repository.*

ECONOMICS & RESEARCH · MEDIA INSIGHTS · THEMATIC ANALYSIS · POLICY RISK

Media Insight Report: Philippine Inflation and Policy Debate

A media insight report turning coverage of the 2026 inflation spike into risks and opportunities for a retail client.

Context. Philippine headline inflation rose from 2.0% in January to a three-year-high 7.2% in April 2026. I wanted to show how media coverage of that spike translates into concrete risks and opportunities for a business client.

Method. I coded coverage into themes (cost-of-living hardship, wage justice versus business costs, and social protection as a shock absorber), read tone qualitatively, and mapped stakeholders for a retail and FMCG client persona.

Findings. Coverage centered on two policy levers: wage hikes of up to P200 a day and expanded social protection, including P43.18B released for cash aid and related programs. For the client, that meant short-term cost and wage pressure, reputational risk around pricing, and medium-term opportunities to align with social protection priorities.

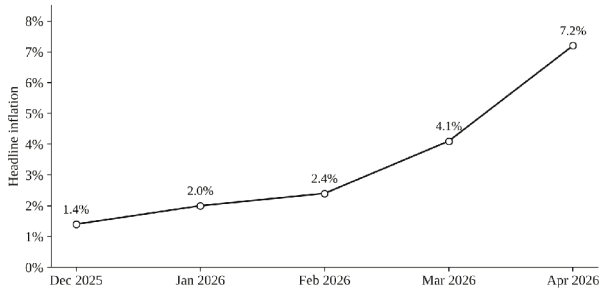


Figure 13. Philippine headline inflation, December 2025 to April 2026. *Source: Philippine Statistics Authority figures compiled in the media insight report.*

Tools: Excel, Media monitoring, Thematic coding, Data visualization
Code: github.com/liancataytayjk-coder/media-audience-engagement-analysis